

The weight of the metal ship, plus the air inside it, is less than the upward force of the water.

Why does the anchor sink while the ship floats?

Floating

A huge metal ship can float because it's full of air. The amount of space the ship takes up weighs less than the equivalent, or same, amount of water.

Water pushes against the weight of the ship. The upward force of the water is greater than the ship's weight, so the ship floats.

The heavy, dense, metal anchor is specially designed to sink in water.

The upward force of the water is less than the weight of the anchor.

Sinking

Objects sink if their weight is greater than the force of the water pushing them upwards. Dense materials, such as metal and stone, usually sink, unless they have air inside them.

Scuba divers use inflatable jackets and weights to move up or down, or stay at the same depth underwater.